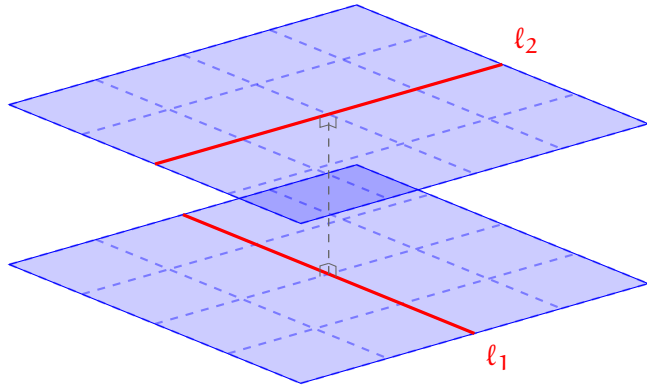


Example 1. Two lines are **skew** if they are not parallel and not intersecting.



Consider the lines with parametrizations:

$$\ell_1 : \vec{r}_1(t) = \langle 1 + t, -2 + 3t, 4 - t \rangle$$

$$\ell_2 : \vec{r}_2(t) = \langle 2t, 3 + t, -1 + 4t \rangle$$

(a) Assess whether the lines are parallel to each other.

Solution. First we find their directions.

$$\vec{v}_1 =$$

$$\vec{v}_2 =$$

Next we attempt to solve:

This has no simultaneous solution λ , hence the directions are not parallel.

(b) Find the distance between the two lines.

Solution. We will take the component of a displacement parallel to a normal.

A normal vector is given by the following.

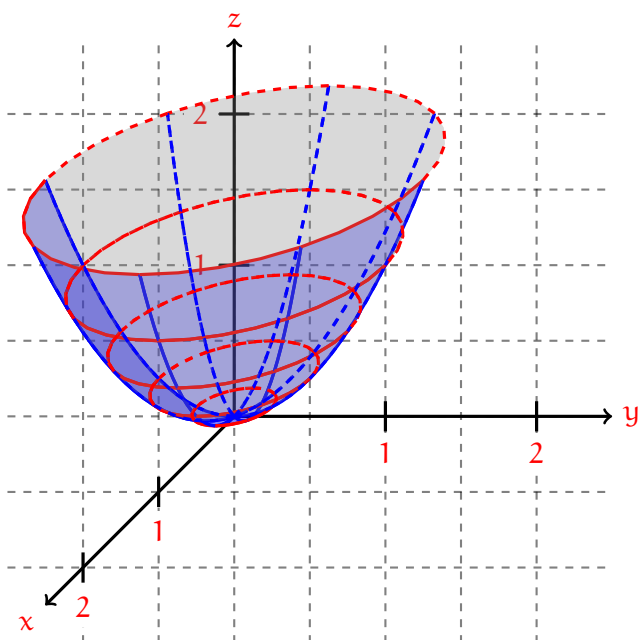
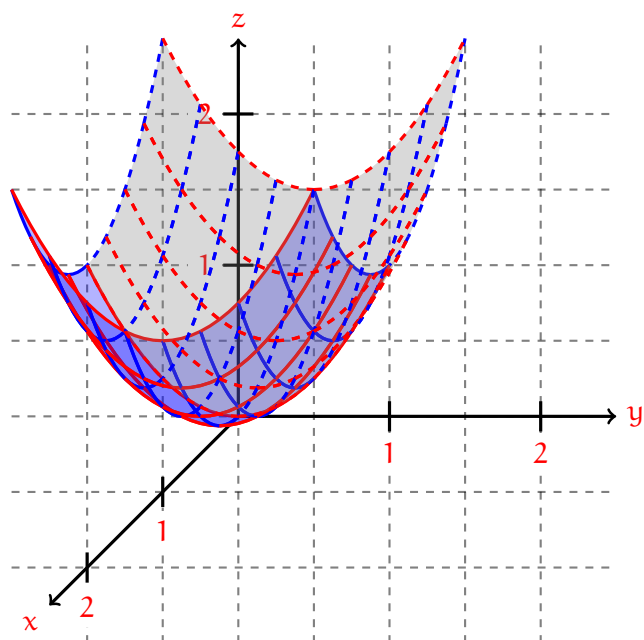
Next, a pair of points, one on each line, and the displacement between them.

To compute the parallel component we use the formula $\text{comp}_{\vec{x}}(\vec{y}) = (\vec{x} \cdot \vec{y}) / \|\vec{x}\|$.

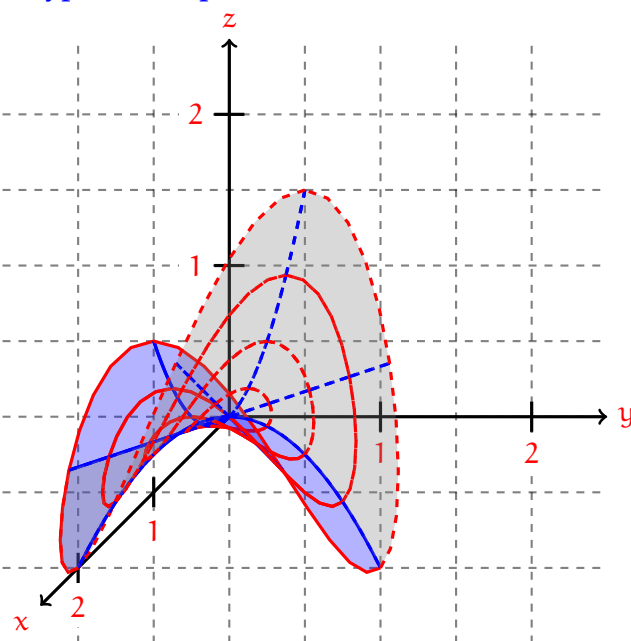
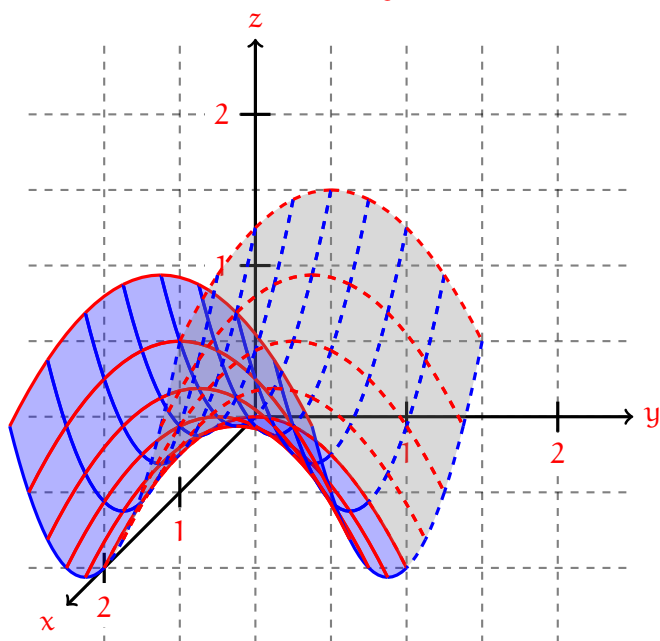
$$\text{distance}(\ell_1, \ell_2) =$$

A. **Paraboloids.** A **quadric surface** is a surface defined by a deg.-2 polynomial.

Among these, first we consider graph of $z = x^2 + y^2$ which we call a **paraboloid**.

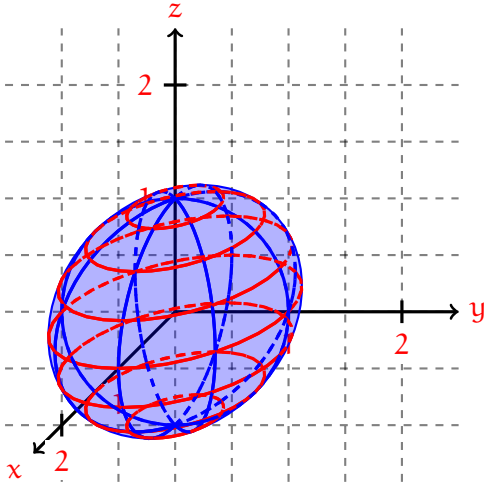


B. **Saddles.** Next $z = x^2 - y^2$ is called a **saddle** or **hyperbolic paraboloid**.



C. **Spheres.** The **unit sphere** consists of points with distance **1** from the origin.

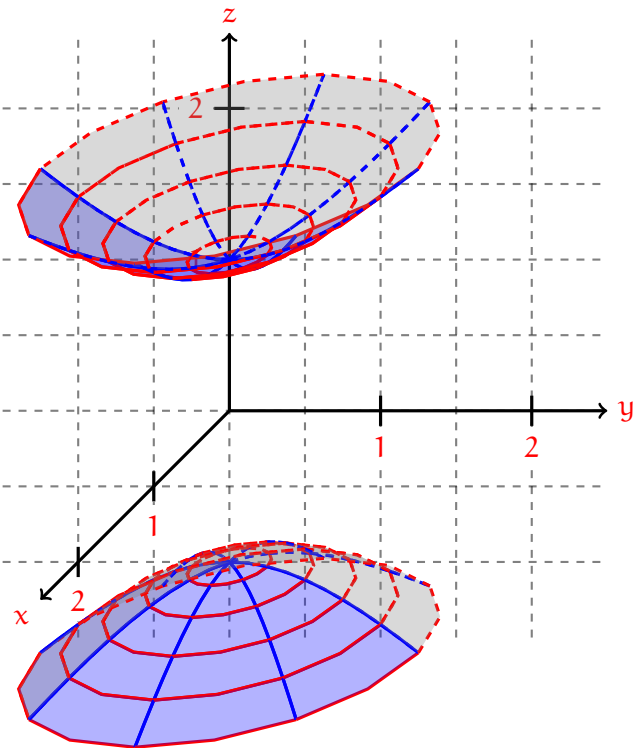
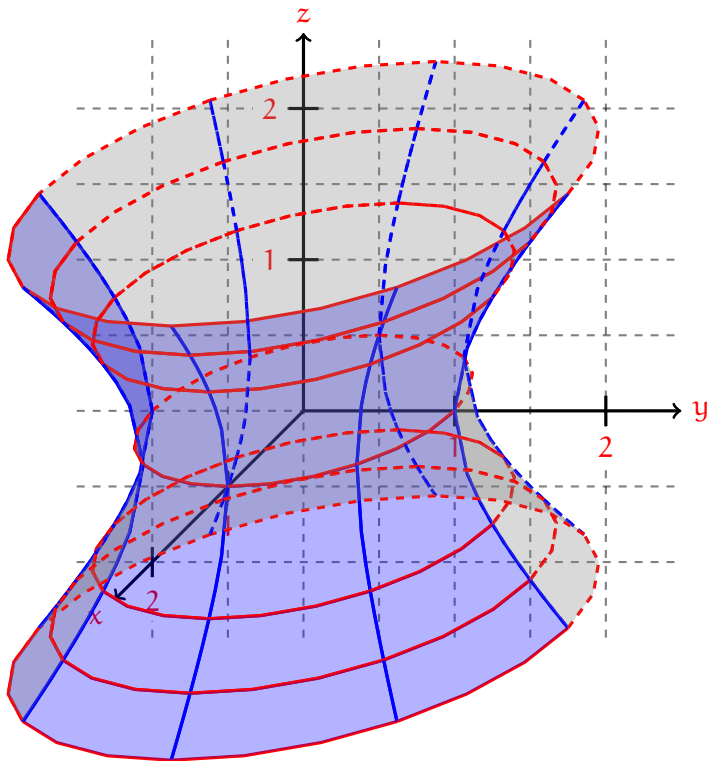
If $\mathbf{X} = (x, y, z)$ is a point on the unit sphere, then its position vector satisfies:



D. **Hyperboloids.** There are two kinds of hyperboloids.

The vertically-opening **one-sheeted hyperboloid** is defined by:

The vertically-opening **two-sheeted hyperboloid** is defined by:



E. **Transformations.** Surfaces can undergo transformations.

Let's begin with the unit sphere $x^2 + y^2 + z^2 = 1$.

From it we first use Desmos3D to visualize a **scaling** to obtain:

$$\left(\frac{x}{2}\right)^2 + \left(\frac{y}{3}\right)^2 + \left(\frac{z}{4}\right)^2 = 1$$

$x \mapsto x/2$ scales the surface by a factor of:

$y \mapsto y/3$ scales the surface by a factor of:

$z \mapsto z/4$ scales the surface by a factor of:

The resulting surface is now called an:

Next we apply a **translation** to obtain:

$$\left(\frac{x-1}{2}\right)^2 + \left(\frac{y-2}{3}\right)^2 + \left(\frac{z-3}{4}\right)^2 = 1$$

$x \mapsto x-1$ translates the surface:

$y \mapsto y-2$ translates the surface:

$z \mapsto z-3$ translates the surface:

Next we apply a **reflection** to obtain:

$$\left(\frac{z-1}{2}\right)^2 + \left(\frac{y-2}{3}\right)^2 + \left(\frac{x-3}{4}\right)^2 = 1$$

swapping $x \leftrightarrow z$ reflects the surface across:

Let us consider the table of standard quadric surfaces below.

Surface:	Standard Equation
paraboloid:	$z = x^2 + y^2$ 
saddle:	$z = x^2 - y^2$ 
ellipsoid—sphere if uniformly scaled:	$x^2 + y^2 + z^2 = 1$ 
1-sheeted hyperboloid:	$x^2 + y^2 - z^2 = 1$ 
2-sheeted hyperboloid	$-x^2 - y^2 + z^2 = 1$ 
double-cone:	$z^2 = x^2 + y^2$ 

You will see the double-cone in discussion section.

For surfaces in this table, scalings, translations, and reflections preserve the: